Incremental Learning With Open-Set Recognition for Remote Sensing Image Scene Classification

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***Abstract*— Image scene classification aiming to assign specific semantic labels for each image is vitally important for the applications of remote sensing (RS) data. In real world, since the observation environment is open and dynamic, RS images are collected sequentially and the numbers of images and classes grow rapidly over time. Most existing scene classification methods are offline learning algorithms, which are inefficient and unscalable for this scenario. In this article, an incremental learning with open-set recognition (ILOSR) framework is proposed for RS image scene classification in the open and dynamic environment, which can identify the unknown classes from a stream of data and learn these new classes incrementally. Specifically, a controllable convex hull-based exemplar selection strategy is designed to address the catastrophic forgetting issue in incremental learning, which can reduce training time and memory footprint effectively. In addition, a new loss function based on prototype learning and uncertainty measurement is proposed for OSR to enhance the interclass discrimination and intraclass compactness of the learned deep features. Experimental results on real RS datasets demonstrate that the proposed method can not only outperform the state-of-the-art approaches on offline classification, incremental learning, and OSR problem separately but also achieve better and more stable performance in the experiments for ILOSR.**

***Index Terms*— Deep learning, incremental learning, open-set recognition (OSR), remote sensing (RS) image scene classification.**

I. Introduction

OVER the past decades, a variety of Earth observation instruments have generated vast amounts of remote sensing (RS) images with different resolutions, which has led to a rapid increase in the demand for intelligent techniques of Earth observations. RS image scene classification plays an important role in Earth observations, and it refers to classifying the scene images into a fixed and meaningful set of land use and land cover classes by interpreting and analyzing the image contents. It has attracted great attention and wide applications in environment monitoring, urban planning, and natural hazards detection.

In recent years, many methods have been proposed for RS image scene classification, which can be generally divided into three categories according to the features: handcrafted feature-based methods, unsupervised feature learning-based methods, and deep feature learning-based methods [1]. In the early studies, due to the limitation of RS image resolutions and feature extraction methods, most scene classification methods are based on handcrafted features, such as color histograms, texture descriptors, scale-invariant feature transform (SIFT) [2], and histogram of oriented gradients (HOG). These handcrafted features are simple and can be rapidly computed, but they have relatively weak representation and discriminative ability. In particular, their performance is limited when dealing with complex scene images. With the improvement of RS image resolutions and the development of feature extraction methods, the handcrafted features are replaced by the features learned based on unsupervised learning. Typical unsupervised feature learning methods include principal component analysis (PCA) [3], autoencoder [4], sparse coding [5], and so on. Chaib *et al.* [6] proposed a feature selection method based on sparse PCA to classify high-resolution satellite images. Cheriyadat [7] utilized a set of functions to convert low-level features into sparse features. In [8], an unsupervised learning framework based on autoencoder was used for RS imagery retrieval. This type of method performs better than handcrafted feature-based approaches, but they do not utilize the label information, which reduces the discriminative ability of features. Currently, the vast majority of the state-of-the-art methods are based on supervised deep learning, especially convolutional neural networks (CNNs), because of its powerful feature learning ability. Cheng *et al.* [9] proposed a rotation-invariant CNN model to enhance the effect of the existing CNN model. Othman *et al.* [10] proposed a domain adaptive network to deal with classification scenarios affected by data shifting problems. Sun *et al.* [11] proposed a gated bidirectional network to integrate the hierarchical feature aggregation and the redundancy information elimination into an end-to-end network. Yu *et al.* [12] proposed an attention generative adversarial network (GAN) for aerial scene

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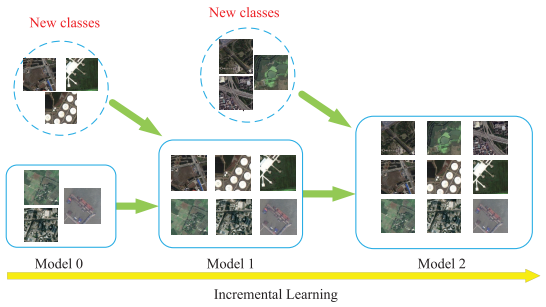


Fig. 1. Incremental learning in RS image scene classification: the model needs to be updated when the new class samples are available.

classification, which combined the attention mechanism with GAN to improve the representation ability of the discriminator. Ma *et al.* [13] proposed a global and local consistent network to deeply explore the useful information of RS images for scene classification. Zheng *et al.* [14] proposed a multitask learning network consisting of a shared feature extractor and multiple task-specific branches, which maintains both generalization and discrimination abilities by learning shared and task-specific features. Besides, Zheng *et al.* [15] proposed a mutual attention inception network for RS visual question answering, which considers the spatial information and semantic word information and introduces attention mechanism and bilinear technique to enhance the features.

However, most existing scene classification methods of RS images are offline/batch learning algorithms, which assume that the observation environment is static with the number of training data and classes fixed at the beginning of learning. In practical applications, RS images are collected sequentially, and the number of images and classes grows rapidly over time. These existing methods are inefficient for the dynamic learning task because they have to retrain the model on both the old and new training samples, which costs a lot of storage memory and computation time. Incremental learning can address the problems of offline or batch learning methods by updating model incrementally when the new data added, as shown in Fig. 1. There are a few incremental learning works [16]–[19], [20], [21] on RS image classification. They can learn new data or new classes incrementally without storing all the training samples for retraining, which have been proven to be effective and efficient to deal with dynamic data streams. In addition, most existing classification methods and incremental learning methods are based on closed-set assumption, that is, all the test samples must belong to the known classes [22]. When encountering samples belonging to an unknown class, the model is forced to classify them as one of the known classes due to the closed-set property. However, in the real-world scenario, the environment is open and the learning task corresponds to an open-set recognition (OSR) problem, where the test samples may belong to the known or the unknown classes, as shown in Fig. 2. The OSR algorithms should have the capability to identify the samples of unseen classes as "unknown" while correctly classifying the samples of known classes. In recent years, many OSR algorithms [23]–[25] have been proposed for the general image classification tasks and only a few OSR works [26], [27] for RS image scene classification. However, they can just identify the samples of unknown classes but cannot learn these new classes incrementally. Thus, when the model encounters samples of these classes again, it just categorizes them as an unknown class and still cannot recognize them separately.

Inspired by the above issues, in this article, we propose an incremental learning with open-set recognition (ILOSR) framework for RS image scene classification, which can identify the unknown classes and learn them incrementally. ILOSR consists of two main parts: the OSR process based on prototype learning and uncertainty measurement, which can identify the samples of unknown classes, and the incremental learning process based on convex hull exemplar selection, which updates the model to learn the new classes incrementally. First, train an initial model based on the current samples and select some exemplars for each class to store in the buffer. Then, identify the unknown classes in the test set based on OSR and manually label them as the new classes. Finally, update the model by using the new class samples and the exemplars of the known classes. Experimental results on real RS images demonstrate the effectiveness of the proposed method. The main contributions of this article are summarized as follows.

1) An ILOSR framework is proposed for RS image scene classification, which can identify the samples of unknown classes and learn these new classes incrementally.
2) A controllable convex hull-based exemplar selection (CHES) strategy is designed to alleviate the catastrophic forgetting problem in incremental learning, which can reduce training time and memory footprint effectively.
3) A new loss function based on prototype learning and uncertainty measurement is proposed for OSR to enhance the interclass discrimination and intraclass compactness of deep features, which makes the learned feature space better separate the known and unknown classes.

II. Related Works

A. Incremental Learning

Incremental learning aims to deal with the continuous information flow in the real-world applications and to learn the new knowledge incrementally from the streamed data without forgetting the previously learned knowledge. For the image classification problem, class-incremental learning means that the classifier should be able to learn new classes incrementally and classify the samples of the new and the old classes accurately. However, the model after being trained on the new class data will suffer from catastrophic forgetting [28], that is, the performance on the old classes drops significantly or even the model forgets the old classes completely. The most direct solution is to use all the data to retrain the model, but it will consume a lot of storage resources and computation time

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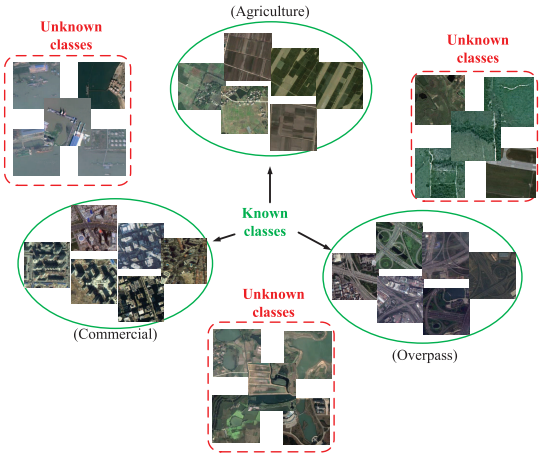


Fig. 2. OSR problem for scene classification: The known class samples are from agriculture, commercial, and overpass. The rest samples are from the unknown classes. The classifier needs to correctly classify the samples of known classes and identify the samples of unknown classes as "unknown."

and it is not applicable to the practical scenario with limited resources.

To address the catastrophic forgetting problem, a variety of incremental learning methods have been proposed, which can be divided into three main categories [29], [30]: 1) replay-based solutions that store some exemplars or generate some samples of previous tasks to retrain the model together with the new data; 2) regularization-based approaches that add soft penalty on the weights to consolidate previous knowledge when learning new data; 3) parameter-isolation methods that dedicate different model parameters to each task by freezing old branches for previous tasks. We will briefly introduce some classical incremental learning approaches as follows. Incremental support vector machine (SVM) [31] only stores the samples that violate Karush-Kuhn-Tucker (KKT) conditions and discards the rest, while VS-OSVM [32] and CHVS [33] only select the convex hull vertices to update the SVM classifier. Incremental classifier and representation learning (iCaRL) [34] picks the nearest neighbors to the average sample of each class as the exemplars set and updates the model using these exemplars of previous classes and the new samples. Elastic weight consolidation (EWC) [35] first establishes the regularization technique to penalize the changes of important parameters, while learning without forgetting (LwF) method [36] introduces the knowledge distillation loss to mitigate forgetting. Expert gate [37] makes a copy of the model for each task and learning an autoencode gate, while PackNet [38] assigns a subset of the model parameters to each incoming task by using iterative pruning and binary masks.

There are only a few studies focusing on incremental learning for RS image scene classification. Li *et al.* [39] proposed a batch mode incremental and active learning framework, which employed two batch selection strategies to alleviate the impact of redundancy. Zhai *et al.* [40] proposed a classification model based on meta-learning, which can learn a classifier from just few-shot samples. Lu *et al.* [41] proposed a lightweight task incremental learning approach, which used a light feature transfer module to realize the alignment of data distributions between different tasks in the feature domain. Also, dual-constraint loss and adversarial learning were introduced to promote the mapping and alignment of data distributions at both the feature level and the semantic level. However, these works assume that the training and test samples must belong to known classes and cannot recognize unknown classes automatically.

B. Open-Set Recognition

Most existing classification models are designed for closed-set problems where the test classes are known during the training stage. As a result, when the test samples are from the unknown classes, the model would classify them as the known classes, which is unreasonable. In the real open environment, the scenario is often encountered that the newly acquired samples may belong to the known or unknown classes. Thus, it is critical to identify the unknown classes automatically. A potential solution for the problem is known as OSR. Scheirer *et al.* [24] defined the OSR problem for the first time based on classifiers with rejection option [42] and proposed the concept of open space risk as follows:

$$R_O(f) = \frac{\int_O f(x)dx}{\int_{S_O} f(x)dx} \tag{1}$$

where O is the positively labeled open space, S_O is a large ball containing O and all the known class samples, and f is a recognition function with $f(x) = 1$ for the known classes and $f(x) = 0$ for the unknown classes.

OSR methods aim to optimize the open space risk and the experimental risk simultaneously, which can be classified into two main types: traditional machine learning-based methods and neural network-based approaches. In [23], an SVM-based method was proposed for OSR, which calibrated the decision scores by the extreme value distribution. Also, two SVM models were used, one for identifying the unknown classes and one for classifying the known classes. Júnior *et al.* [43] proposed an OSR model based on nearest neighbors, and the neighbor similarity was used as the decision scores for open-set identification. In [24], a novel "1-vs-set machine" was proposed, which used one-class or binary SVM to shape a decision space. With the great achievements of neural networks in computer vision tasks, many neural network-based methods are proposed for OSR problem. Neal *et al.* [44] proposed a data augmentation method, which used a GAN to generate the fake images based on the known classes. These fake images are treated as the unknown class samples and combined with the known class samples to train a neural network. In [45], a new concept reciprocal point was proposed, which represents the latent representation in feature space associated with each class. The new proposed learning framework is called reciprocal point learning (RPL), which increases the compactness and discrimination of the learned representations. Yang *et al.* [46] proposed a convolutional prototype network (CPN), which learned representations by CNN and replaced the softmax with an open-world oriented and human-like prototype model.

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There are only a few studies on OSR for RS image scene classification. Zhang *et al.* [26] proposed an open-set domain adaptation network (OSDANet) by designing a feature generator as well as a classifier, which was learned in an adversarial way. Adayel *et al.* [27] *et al.* proposed an OSR method based on adversarial learning and Pareto ranking, which utilized min–max entropy optimization to leverage the distribution discrepancy between the source and target domains. However, the two works can just identify the samples of unknown classes and cannot learn these new classes incrementally.

Although there are many incremental learning and OSR methods, most of the existing incremental learning algorithms are still based on the closed-set assumption, while the open-set algorithms do not take the incremental learning into account. There are few works to consider open-set problem under incremental learning setting, which is commonly encountered in real-world applications. Dang *et al.* [47] proposed an OSmIL method to solve the OSR problem in the incremental learning scenario for automatic target recognition (ATR), which can constantly identify and learn new unknown classes. However, this method is designed for synthetic aperture radar (SAR) ATR and cannot be extended to other classification tasks with complex image information directly, such as RS image scene classification.

Algorithm 1 Framework of ILOSR

Require:
$X^1, \ldots, X^K$: training sample sets of the initial classes;
X^{test}: test sample sets including some new classes;
H: maximum number of exemplars per class.

1: Train the CNN to obtain a feature mapping function and a SVM classifier:
$\psi_\theta \longleftarrow CNN(X^1, \ldots, X^K)$;
$SVM \longleftarrow \text{SVM}(\psi_\theta(X^1), \ldots, \psi_\theta(X^K))$
2: Select exemplar sets:
$S^1, \ldots, S^K \leftarrow CHES(\psi_\theta(X^1), \ldots, \psi_\theta(X^K); H)$;
3: Detect the unknown samples and label them:
$X^{new} = OpenSetRecognition(X^{test}, \psi_\theta)$;
$X^{K+1}, \ldots, X^{K+s} \leftarrow X^{new}$;
4: **if** $X^{K+1}, \ldots, X^{K+s}$ are available **then**
5: Update the feature mapping function:
$\psi_\theta \longleftarrow CNN(S^1, \ldots, S^K, X^{K+1}, \ldots, X^{K+s})$;
6: Update the OSR model;
7: Select exemplars of the new classes and update exemplars of the previous classes:
$S^1, \ldots, S^{K+s} \leftarrow CHES(\psi_\theta(S^1), \ldots, \psi_\theta(S^K), \psi_\theta(X^{K+1}), \ldots, \psi_\theta(X^{K+s}); H)$;
8: **end if**

III. Proposed Method

In this section, we present the framework and main components of the proposed ILOSR method. Section III-A gives an overview of ILOSR with the framework shown in Fig. 3(a). Section III-B introduces the proposed incremental learning algorithm based on CHES, and Section III-C provides the proposed OSR algorithm based on prototype learning.

A. Framework of ILOSR

Most existing incremental learning algorithms assume that the test samples must belong to known classes and cannot recognize unknown classes automatically, while most existing OSR algorithms need to be retrained from scratch when new training samples and target classes added. Therefore, we propose ILOSR that combines the capabilities and advantages of incremental learning and OSR algorithms. ILOSR can not only accomplish incremental learning and OSR separately but also identify the unknown classes while learning the new classes without retraining from scratch.

Fig. 3(a) presents the framework of ILOSR and Fig. 3(b) and (c) shows the two main components of ILOSR. Given the initial sample sets $X^1, X^2, \ldots, X^K$, all samples in the set $X^k = \{x_1^k, x_2^k, \ldots, x_n^k\}$ belong to class $k \in \{1, \ldots, K\}$, where K is the initial number of classes and n is the number of samples in each set. The training of CNN model starts with the initial K classes and some exemplars for each class are selected to store. When test samples, including known and unknown classes, arrive, the current OSR model can classify known samples and identify unknown samples. These unknown samples, labeled manually and combined with the selected exemplars, are used for incremental learning in the next round. As shown in Fig. 3(b), exemplars are obtained based on the convex hull vertices selection and the closed-set classification utilizes the SVM classifier. As shown in Fig. 3(c), OSR is based on prototype learning and one-class SVM, which can establish the close decision boundary for each known class. If a test sample is within a decision boundary, it will be assigned to the corresponding class. If all decision boundaries reject it, it belongs to the unknown classes.

Algorithm 1 shows more details about the framework of ILOSR, which includes four main steps: 1) train a CNN using the samples of initial classes to map images into the feature space: $\psi_\theta{:}x \rightarrow \mathbb{R}^d$, where θ is the parameters of CNN and an SVM classifier; 2) choose some representative samples for each old class based on convex hull exemplars selection; 3) detect the unknown classes in the newly arrived test samples using the OSR model automatically and then label them as new classes manually; and 4) update the CNN and OSR models by using the new labeled samples and the stored exemplars and update the exemplars of both old and new classes.

B. Incremental Learning Based on Convex Hull Exemplar Selection

For an incremental learning model, selecting suitable exemplars is critical because exemplars can not only reduce training time and storage resources but also can help to address catastrophic forgetting. In this section, we introduce the convex hull vertex-based exemplars selection technique to choose the most representative samples for each class. Fig. 4 shows the geometrical property of SVM, that is, the solution of SVM is equivalent to finding the minimum distance between two convex hulls formed by the two class samples for linearly separable problem and the SVM classifier only relates to the

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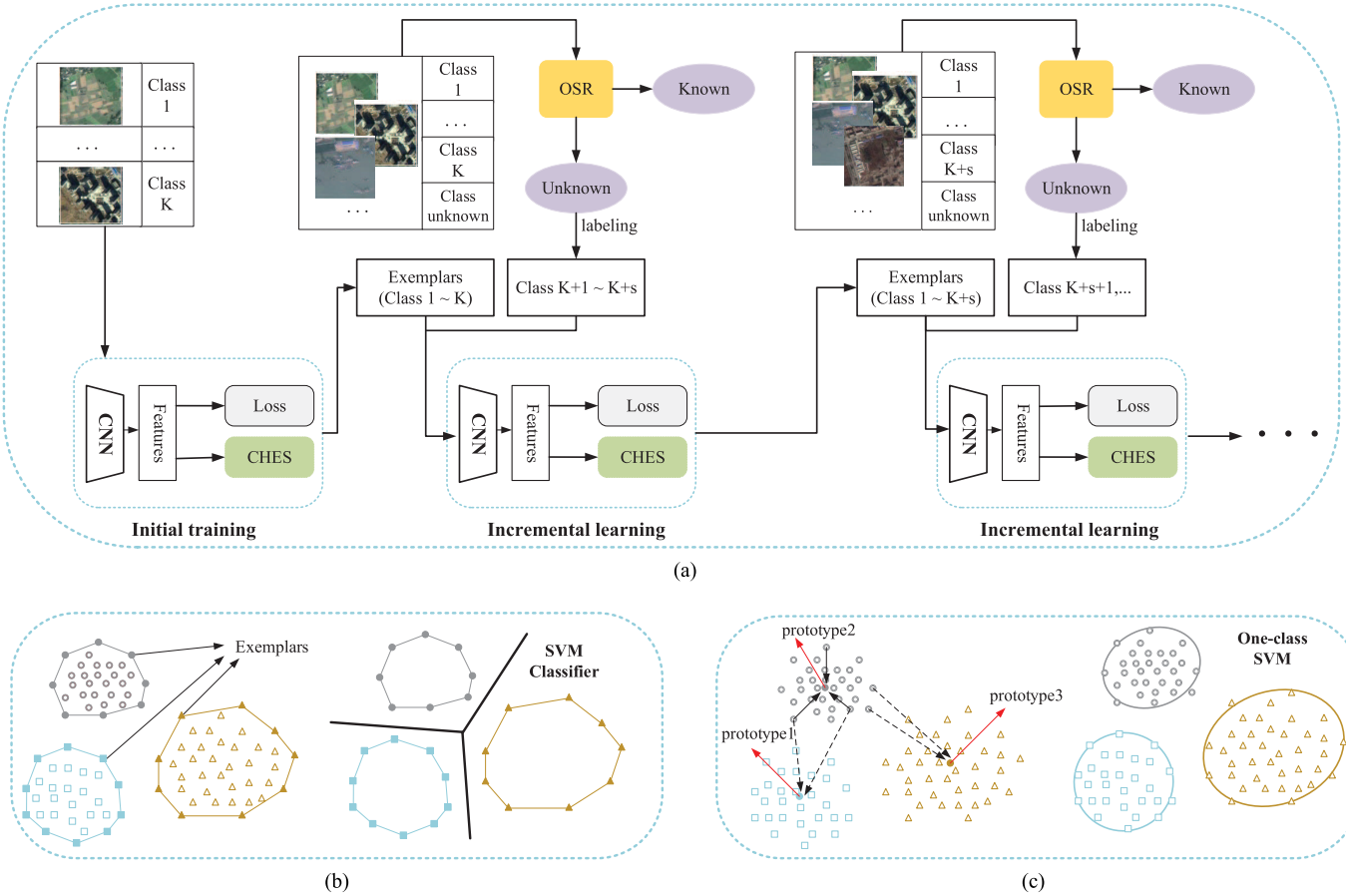


Fig. 3. Overview of ILOSR. (a) Framework of ILOSR. (b) CHES and SVM-based closed-set classification. (c) OSR based on prototype learning and one-class SVM.

convex hull vertices [33]. Based on this point, a small number of skeleton samples can be selected as exemplars for each class in the feature space, which can form an approximate convex hull covering the whole samples. Wang *et al.* [32] proposed an online SVM classifier based on approximate convex hull by solving a quadratic optimization problem to compute the distance between a sample and a convex hull, which is time-consuming. Ding *et al.* [33] proposed a fast convex hull vertices selection algorithm for online classification, which converts the quadratic optimization problem into a linear equation problem with much lower computational complexity.

Given a set $S = \{x_i\}_{i=1}^{d}$, $\psi_\theta(x_i) \subset \mathbb{R}^d$, the convex hull of S is defined by $\text{conv}(S) = \{\sum_{i=1}^{d}\alpha_i\psi_\theta(x_i)|x_i \in S,\ \sum_{i=1}^{d}\alpha_i = 1, \alpha_i > 0, i = 1,\ldots,d\}$, and the hyperplane composed of S is defined by $H = \{\sum_{i=1}^{d}\alpha_i\psi_\theta(x_i)|x_i \in P,\ \sum_{i=1}^{d}\alpha_i = 1, \alpha_i \subset \mathbb{R}^d, i = 1,\ldots,d\}$. d_c represents the distance between a sample x and $\text{conv}(S)$ and d_p represents the distance between a sample x and H. If the projection of x onto H is in $\text{conv}(S)$, d_c can be computed as follows:

$$d_c(x, \text{conv}(S)) = d_p(x, H) = \left\|\beta^T(\psi_\theta(x) - \psi_\theta(x_1))\right\|_1 / \|\beta\|_2 \tag{2}$$

where $\|\cdot\|_1$ denotes the l_1-norm, $\|\cdot\|_2$ denotes the l_2-norm, and β is the normal vector of the hyperplane H satisfying the following conditions:

$$\beta^T\psi_\theta(x_i) - 1 = 0, \quad i = 1,\ldots,d. \tag{3}$$

If the projection of x onto H is not in $\text{conv}(S)$, d_c can be calculated as follows:

$$d_c(x, \text{conv}(S)) = \min_i \|\psi_\theta(x) - \psi_\theta(x_i)\|_2. \tag{4}$$

Thus, the distance between a sample and the current convex hull can be replaced by computing d_p and d_c in (2)–(4).

Notice that in [33], the rule of convex hull vertex selection is based on the maximum distance between samples and their corresponding subconvex hulls, that is, after choosing $d + 1$

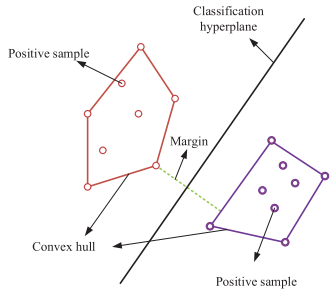


Fig. 4. Geometric feature of SVM [32].

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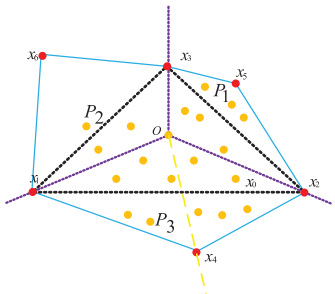


Fig. 5. CHES method in 2-D space.

vertices, the sample with the largest distance to its subconvex hull is selected as the $(d+2)$th vertex. However, this rule is sensitive to outliers and cannot guarantee to cover as many samples as possible because the part with the largest distance does not necessarily contain the most samples. Therefore, when the number of selected samples is limited, these selected vertices may not be the most representative since the convex hull formed by them may not be the one that contains the most samples. Based on this point, we develop a controllable CHES method, which first finds the subconvex hull that includes the most samples and then selects the furthest point from the corresponding part as a vertex. This rule can ensure that the convex hull composed of these selected points covers the most samples. Thus, when the algorithm needs a certain number of exemplars for each class, the selected vertices by the proposed method are more representative, while the rule in [33] is sensitive to outliers, as shown in Fig. 5. In addition, the method in [33] requires to compute the distances between all the samples in each part and their corresponding subconvex hull, which is time-consuming. In contrast, the proposed method only needs to compute the distances between the points in the part containing the most samples and the corresponding subconvex hull, which costs much less time. Finally, the method in [33] is designed for online learning, which is modified and introduced to incremental learning in this article.

Fig. 5 shows the process of CHES method in the 2-D feature space. First, $d+1$ points should be selected for a d-simplex. Thus, we need to choose three points in the 2-D space with the following steps: randomly select a point and denote it as x_0; find the furthest point to x_0 and denote it as x_1; select the furthest point to x_1 and denote it as x_2; and find the furthest point to the straight line that passes through x_1 and x_2 and denote it as x_3. Then, the set $S=\{x_1,x_2,x_3\}$ is the initial vertex set of a two-simplex. O is the mean of the set S. All points outside of conv(S) are divided into three parts by Ox_1, Ox_2, and Ox_3, denoted by P_1, P_2, and P_3, and $S_1=\{O,x_2,x_3\}$, $S_2=\{O,x_1,x_3\}$, and $S_3=\{O,x_1,x_2\}$ denote the subconvex hull of each part, respectively. The above process corresponds to steps 1–3 of Algorithm 2. Then, find the part P_3 that contains the most points and select the furthest point x_4 to the current convex hull. The part P_3 is divided into two small parts and x_4 is added to S. At this time, the feature space can be divided into four parts. Repeat the previous step to find the next vertex until the number of vertices reaches a given value or all points are contained by the convex hull. In Fig. 5, $S=\{x_1,x_2,x_3,x_4,x_5,x_6\}$ is the final set of exemplars. More details are presented in Algorithm 2, where I is the index of the divided parts, M is the number of the divided parts, P_{oc} represents the set of the divided parts, P_i is the ith part of P_{oc}, and S_i denotes the set of the vertices of the corresponding facet of P_i, $B_i(i=1,\ldots,d)$ represent the newly divided parts of an old part, and $E_i(i=1,\ldots,d)$ are the sets of the vertices of the newly generated facets. From Fig. 5, it can be seen that the exemplar selection order of our method is $x_1\rightarrow x_2\rightarrow x_3\rightarrow x_4\rightarrow x_5\rightarrow x_6$, while the selection order in [33] is $x_1\rightarrow x_2\rightarrow x_3\rightarrow x_6\rightarrow x_4\rightarrow x_5$ with x_6 being an outlier. If selecting four exemplars, the convex hull $conv(x_1,x_2,x_3,x_4)$ given by our method covers more samples and provides more information than $conv(x_1,x_2,x_3,x_6)$ given in [33].

When the new incoming training data include the samples of both the old and new classes, the exemplar sets of the old classes need to be updated, while that of the new classes should be created. It means that some samples in the old exemplar sets will be replaced by the new ones. In addition, since CHES is based on the geometric feature of SVM, ILOSR adopts the SVM classifier to classify instances in the closed-set scenario, which can provide a prior category for OSR and present good generalization performance.

C. OSR Based on Prototype Learning

In this section, we introduce a new OSR method based on prototype learning and uncertainty measurement. Traditional CNNs consist of two main parts: 1) feature extractor $\psi_\theta(x)$ mapping the raw images into the feature space and 2) a linear classifier $C(\psi_\theta(x);\theta_c)$ performed by the last fully connected layer to output a vector whose length is the number of target classes. The softmax operation is used to scale the output such that the sum of posterior probabilities of all known classes is 1 during training

$$p(y|x)=\frac{e^{w_y\cdot\psi_\theta(x)+b_y}}{\sum_{k=1}^{K}e^{w_k\cdot\psi_\theta(x)+b_k}},\quad \sum_{k=1}^{K}p(k|x)=1 \tag{5}$$

where w_k and b_k are the weights and bias parameters of class k, respectively. The cross-entropy (CE) loss is usually used to train the model, which is defined as follows:

$$\mathcal{L}_{\text{ce}}=-\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}\mathbb{I}_i(k)\left[\log p_i(k)\right] \tag{6}$$

where $\mathbb{I}_i$ is an indicator function for label i (i.e., one-hot encoded vector), p_i is a predicted probability score vector with $p_i(k)$ being the probability of the ith sample from the kth class, K is the current number of classes, and N is the number of samples in each minibatch. CNN models trained by (6) assume that the whole feature space is for the known classes without considering the unknown classes, and they do not employ any measures to manage the open space risk. This is the reason why the classifier gives poor performance while dealing with unknown classes.

To address the above issue, we design a novel CE loss function based on uncertainty distance measurement and prototype

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Algorithm 2 CHES

Input:

$X^y = \{x_1, \ldots, x_n\}$, $\psi_\theta(x) \subset \mathbb{R}^d$: training samples of class y;

H: maximum number of exemplars per class;

ψ_θ: the feature mapping function.

Output:

exemplar set S

Randomly select a sample x_i as x_0. Let $x_{j_0} = \text{argmax}_{x \in X^y} \|\psi_\theta(x_0) - \psi_\theta(x)\|_2$ and $x_{j_1} = \text{argmax}_{x \in X^y} \left\|\psi_\theta\left(x_{j_0}\right) - \psi_\theta(x)\right\|_2$. Initialize $S = \left\{x_{j_0}, x_{j_1}\right\}$, $I = \{1, \ldots, d\}$ and $M = d + 1$;

Select $d + 1$ samples as the vertices of a d-simplex in $\mathbb{R}^d$ by using Algorithm 1 of [32]. $S = \{x_i\}_{i=1}^{d+1}$ is the set of selected samples;

Divide the samples into several parts by using the Constrained Data Partion Algorithm 2 of [32] and obtain the sets P_{oc} and S_{oc}, where $P_{oc} = \{P_i\}_{i=1}^{d+1}$ and P_i is the ith divided part of X^y obtained by Algorithm 2 of [32] and $S_{oc} = \{S_i\}_{i=1}^{d+1}$ with $S_i = \{x_k\}_{k=1k\neq i}^{d+1}$;

Compute the number of samples of each part, $N_P = \{N_1, \ldots, N_{d+1}\}$, and the number of S, denoted by N_S;

while max $N_P > 0$ *or* $N_S < H$ **do**

  Let $j_0 = \text{argmax} N_P$ and find the sample $x_{j_0}^* \in P_{j_0}$, where P_{j_0} contains the most samples and $x_{j_0}^*$ is the furthest from the convex hull $\text{conv}\left(S_{j_0}\right)$ by using Eq. (2). $x_{j_0}^{(d+1)} = x_{j_0}^*$;

  $S = S \cup \left\{x_{j_0}\right\}$, $I = I \setminus \{j_0\}$;

  Let $E_i = \left\{x_{j_0}^{(t)}\right\}_{t=1t\neq i}^{d+1}$ for $i = 1, \ldots, d$;

  Divide P_{j_0} into d parts and obtain $B_i, i = 1, \ldots, d$, where B_i is the i_{th} part;

  Let $P_{M+i} = B_i, S_{M+i} = E_i \ for \ i = 1, \ldots, d$;

  Let $I = I \cup \{M + 1, \ldots, M + d\}, M = M + d$;

  Compute the number of samples of each part, $N_P = \{N_1, \ldots, N_I\}$, and the number of S, denoted by N_S;

end while

Obtain the exemplar set S.

learning. We set a prototype c_k for each class in feature space, and the Euclidean distance between a sample and the prototype is used to measure the uncertainty of the sample belonging to the corresponding class. Thus, the probability of a sample x belonging to class k is proportional to the distance between $\psi_\theta(x)$ and the corresponding c_k

$$p(y = k|x) \propto -d(\psi_\theta(x), c_k) \tag{7}$$

where $d(\psi_\theta(x), c_k) = \|\psi_\theta(x) - c_k\|_2^2$ represents the distance. The prototypes $\{c_k\}$ can be regarded as the class center in deep feature space. It is a learnable vector that is initialized at the beginning of training and then continually updated, as shown in Algorithm 3. Furthermore, we define the probability that a sample x belongs to the class k as follows:

$$p(y = k|x) = \frac{e^{-\gamma d(\psi_\theta(x), c_k)}}{\sum_{i=1}^{K} e^{-\gamma d(\psi_\theta(x), c_i)}} \tag{8}$$

where γ is a hyperparameter that controls the hardness of probability. Based on the probability $p(y = k|x)$, we can

Algorithm 3 Feature Learning Algorithm

Require:

$X^1, \ldots, X^K$: training samples of initial classes;

θ, c, R: initialized parameters;

μ, μ_c, μ_R: learning rate for θ, c, R;

λ_R, γ: hyperparameters;

t: the number of training iteration.

Ensure:

the parameters θ, c and R.

while not converge **do**

  $t \leftarrow t + 1$

  Compute the joint loss by $\mathcal{L}^t = \mathcal{L}_{CL}^t + \lambda_R \mathcal{L}_R^t$.

  Compute the backpropagation error $\frac{\partial \mathcal{L}^t}{\partial x_i^t}$ for each i by $\frac{\partial \mathcal{L}^t}{\partial x_i^t} = \frac{\partial \mathcal{L}_{CL}^t}{\partial x_i^t} + \lambda_R \frac{\partial \mathcal{L}_R^t}{\partial x_i^t}$.

  Update the parameters θ by $\theta^{t+1} = \theta^t - \mu^t \cdot \frac{\partial \mathcal{L}^t}{\partial \theta^t} = \theta^t - \mu^t \cdot \left(\frac{\partial \mathcal{L}_{CL}^t}{\partial \theta^t} + \lambda_R \cdot \frac{\partial \mathcal{L}_R^t}{\partial \theta^t}\right)$.

  Update the parameters c by $c^{t+1} = c^t - \mu_c^t \cdot \frac{\partial \mathcal{L}^t}{\partial c^t}$ according to (12).

  Update the parameters R by $R^{t+1} = R^t - \mu_R^t \cdot \frac{\partial \mathcal{L}^t}{\partial R^t}$ according to (13).

end while

Algorithm 4 OSR

Require:

$X^1, \ldots, X^k, \ldots, X^K$: training samples of initial classes with X^k being the samples of class k;

$X = \{x_1, \ldots, x_m\}$: test samples of the known and unknown classes;

ψ_θ: the feature mapping function trained in Algorithm 1;

SVM: SVM classifier learned in Algorithm 1.

for $i = 0; i < K; i++$ **do**

  $M_i \longleftarrow \text{one} - \text{classSVM}\left(\psi_\theta\left(X^i\right)\right)$;

end for

for $x_m \in X$ **do**

  provide a priori category $i \leftarrow SVM(\psi_\theta(x_m))$

  if $M_i(\psi_\theta(x_m)) = 1$ **then**

    predict x_m as known class with label i;

  else

    predict x_m as unknown class;

  end if

end for

redefine the CE loss function

$$\mathcal{L}_{\text{CL}} = -\sum_{i=1}^{N} \sum_{k=1}^{K} \mathbb{I}_i(k) \left[\log p_i(y = k|x)\right]. \tag{9}$$

According to (7)–(9), it can be concluded that minimizing this loss can decrease the distances between samples and correct prototypes of their true classes and increase the distances between samples and incorrect prototypes, as shown in Fig. 3(c). It means that $\mathcal{L}_{\text{CL}}$ can manage the empirical risk.

In addition, to address the OSR problem, it is also critical to manage the open space risk. Obviously, it is impossible to do this by constraining the boundaries of the open space because a large number of unknown classes and samples are in the

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open space. Thus, it is a feasible way to constrain the space occupied by the known classes. We introduce a regularization term to constrain the distance between samples of class k and their corresponding prototypes c_k and let samples of the same class be on a hypersphere as follows:

$$\mathcal{L}_R = \sum_{k=1}^{K} \sum_{i=1, y_i \neq k}^{N} \|d(\psi_\theta(x_i), c_k) - R_k\|_2^2 \tag{10}$$

where R_k is the radius of class k. It is not a fixed value, but a learnable parameter. Minimizing the function in (10) can help to manage the open space risk. Finally, the total loss function is defined as follows:

$$\mathcal{L}(\psi_\theta, c, R) = \mathcal{L}_{\mathrm{CL}}(\psi_\theta, c) + \lambda_R \mathcal{L}_R(\psi_\theta, c, R) \tag{11}$$

where λ_R is a hyperparameter to balance the two terms.

The prototype for each class c_k is a learnable vector and it is continually optimized with the number of iterations increasing by $c^{t+1} = c^t - \mu_c^t \cdot \frac{\partial \mathcal{L}^t}{\partial c^t}$. The gradient of the total loss in (11) with respect to c_k is

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial c_k} = {} & 2\gamma \sum_{i=1, y^i=k}^{N} \left(c_k - \psi_\theta\left(x^i\right)\right) \\ & + 2\gamma \sum_{i=1}^{N} \left[\left(\psi_\theta\left(x^i\right) - c_k\right) \frac{e^{-\gamma d\left(\psi_\theta\left(x^i\right), c_k\right)}}{\sum_{j=1}^{K} e^{-\gamma d\left(\psi_\theta\left(x^i\right), c_j\right)}} \right. \\ & \left. - 4\lambda_R \sum_{i=1, y^i=k}^{N} \left[d\left(\psi_\theta\left(x^i\right), c_k\right) - R_k\right] \left(\psi_\theta\left(x^i\right) - c_k\right) \right]\end{aligned} \tag{12}$$

where x^i is the ith sample and y^i is the label of sample x^i. Similarly, the radius for each class R_k is also a learnable parameter and it is updated according to $R^{t+1} = R^t - \mu_R^t \cdot (\partial \mathcal{L}^t / \partial R^t)$. The gradient of the total loss with respect to R_k is

$$\frac{\partial \mathcal{L}}{\partial R_k} = 2\lambda_R \sum_{i=1, y^i=k}^{N} \left[R_k - \left\|\psi_\theta\left(x^i\right) - c_k\right\|^2\right]. \tag{13}$$

Features learned based on the loss in (11) present larger interclass discrimination and better intraclass compactness. Algorithm 3 presents the details of feature learning.

Note that the proposed loss function is different from that in [45]. First, the specific formulas and variable meanings of the two loss functions are different. The loss in [45] uses reciprocal point to represent the features of unknown classes with each class corresponding to one or several reciprocal points. In this article, we utilize prototype to denote the center of each known class in feature space. Second, the implications of the distance-based CE term in the two loss functions are different. In [45], this term is based on the distance between samples and the reciprocal points and aims to keep a sample away from the reciprocal points. In this article, this term is based on the distance between samples and the prototypes and aims to keep a sample closer to its corresponding prototype and far away from the other prototypes. Third, the effects of the regularization term in the two loss functions are different. In [45], the regularization term aims to make the distance close to the margin in order to separate the samples of known classes from the reciprocal points indicating unknown classes. In this article, the regularization term attempts to constrain the space of the known classes, which makes the samples of each class distribute in a hypersphere with the rest space for unknown classes.

In the OSR problem, the closed decision boundaries need to be established. Most methods utilize the probability or distance threshold to determine whether a sample belongs to a certain class. However, it is usually difficult to set a suitable threshold. In our work, a combined classification method of SVM and one-class SVM is proposed for OSR. Specifically, SVM is utilized to preclassify the test samples and output their prior labels, while one-class SVM is employed to establish the closed decision boundaries based on the results of SVM. This is because SVM and one-class SVM have similar principles and the decision boundaries established by one-class SVM are located in the subspace divided by SVM. In contrast, if only SVM is used, it cannot identify unknown classes since the classification subspace is semiopen. If only one-class SVM is used, each test sample must be compared with all the decision boundaries one by one, which is time-consuming. Besides, it is unable to identify a sample if it is in the overlap part of two decision boundaries. If using the proposed method, the above problems can be addressed. The process of OSR is presented in Algorithm 4. First, SVM and one-class SVM classifiers are learned based on the training samples. Next, for a test sample x, SVM provides *a priori* class i, and then, one-class SVM judges whether x is in the boundary of ith class. If so, predict that x belongs to class i; otherwise, it belongs to the unknown class.

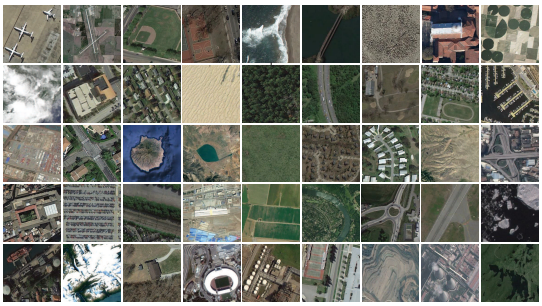
IV. Experiments and Results

A. Dataset and Experiment Setting

To evaluate the performance of ILOSR, we use two RS image datasets: NWPU-RESISC45 [1] and SIRI-WHU [48]. NWPU-RESISC45 dataset is collected by Northwestern Polytechnical University (NWPU) and consists of 45 scene classes, as shown in Fig. 6(a). It contains 31 500 images in total with each class including 700 images. The image size is 256 × 256, and the spatial resolution is from 30 to 0.2 m for most scene classes, while the classes of island, lake, mountain, and snowberg have lower spatial resolutions. SIRI-WHU dataset is collected by the RS image extraction and analysis team of Wuhan university and consists of 12 scene classes, as shown in Fig. 6(b). It contains 2400 images in total with each class including 200 images. Its image size is 200 × 200, and the spatial resolution is 2 m.

For offline RS image scene classification, 10% or 20% samples are randomly selected as training set with the rest as the test set. For incremental learning, the two datasets are split into training and test sets in the ratio of 4:1. For the SIRI-WHU dataset, since the number of samples in each class is small, this dataset is augmented by horizontal flipping, rotating 90°, 180°, and 270° operations. The numbers of the training and test sets of the two datasets are shown in Table I.

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(a)



(b)

Fig. 6. Some example images. (a) NWPU-RESISC45 dataset (45 classes). According to the order from left to right and top to bottom: airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circle farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snowberg, sparse residential, stadium, storage tank, tennis tank, terrace, thermal power station, and wetland. (b) SIRI-WHU dataset (12 classes): agriculture, commercial, harbor, river, water, meadow, overpass, park, pond, residential, idle land, and industrial.

TABLE I
NUMBERS OF SAMPLES FOR NWPU-RESISC45 AND SIRI-WHU DATASETS

Datasets	Training Data	Augmented Training Data	Test Data
NWPU-RESISC45	25200	25200	6300
SIRI-WHU	1920	15360	480

We utilize the ResNet50 and ResNet18 networks as backbones for offline and incremental learning tasks, respectively. The fully connected layer of the two networks is modified to obtain the 200-D features. An Adam optimizer is used and the batch size is 32 with 60 epochs. The learning rate for the radius R is set as 0.01, and the learning rate for the other parameters is set as 0.0001 and reduces to 1/2 of the previous learning rate after every 15 epochs. γ and λ_R are set as 1.0 and 0.05, respectively. The prototype and radius of each class are initialized with ones. The following standard multiclass classification accuracy is used to evaluate the quantitative performance:

$$\text{Accuracy} = \frac{\sum_{i=1}^{K} \text{TP}_i}{\sum_{i=1}^{K} \text{TP}_i + \text{FN}_i} \tag{14}$$

where TP_i (true positive) represents the number of samples that are predicted to be the class i and actually belong to the class i. FN_i (false negative) represents the number of samples that are predicted to be the other classes but actually belong to the class i. The values of evolution metrics averaged over five randomized trials are reported in the following experiments.

TABLE II
EFFECTS OF EXEMPLAR SELECTION STRATEGY AND LOSS FUNCTION IN INCREMENTAL LEARNING ON NWPU-RESISC45

number of classes	15	20	25	30	35	40	45
NS + our loss	**0.9809**	0.9460	0.9305	0.9031	0.9051	0.9003	0.8950
MES + our loss	0.9804	0.9653	0.9563	0.9359	0.9312	0.9262	0.9214
CHES + our loss	**0.9809**	**0.9775**	**0.9680**	**0.9497**	**0.9422**	**0.9382**	**0.9347**
CHES + CE loss	0.9800	0.9710	0.9560	0.9383	0.9259	0.9246	0.9211

TABLE III
EFFECTS OF THE NUMBER OF SELECTED EXEMPLARS IN INCREMENTAL LEARNING

Number of exemplars per class	65	130	195	260	325	560 (all)
accuracy over 45 classes	0.8592	0.9106	0.9266	0.9347	0.9354	0.9466
training time per epoch (/second)	40.14	58.12	77.11	97.02	116.20	191.41

B. Ablation Study

The proposed method consists of two main components: the CHES strategy and the new loss function. In this section, the effect of each module is analyzed by comparing with other exemplar selection approaches and the standard CE loss. In addition, the impact of the number of selected exemplars is also given in terms of the accuracy and training time. These experiments are carried out on NWPU-RESISC45 in the incremental learning scenario: the first 15 classes are set as the initial classes and five classes are added each learning round according to the order in Fig. 6(a). ResNet18 network is employed to learn features and a one-vs-one SVM with linear kernel and $C = 0.0081$ is used as the classifier. The average accuracy in (14) is used to evaluate the quantitative performance.

Table II shows the ablation study results of the following four experiments.

1) *Nearest Selection (NS) [34] + Our Loss:* Select m samples nearest to the class mean in the feature space as exemplars and use the proposed loss function in (11) to train the network model.
2) *Max-Entropy Selection (MES) [49] + Our Loss:* Select m samples with the highest entropy for each class and use our loss.
3) *CHES + Our Loss:* Select m samples by convex hull exemplar selection strategy and use our loss, which is the proposed method.
4) *CHES + CE Loss:* Select m samples by convex hull exemplar selection strategy and use the CE loss.

It can be seen that under the same setting of loss function, the proposed CHES method outperforms the other exemplar selection methods and it achieves 3.91% higher accuracy than

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TABLE IV
RS IMAGE SCENE CLASSIFICATION ON THE NWPU-RESISC45 DATASET

Methods	Training ratios	
	10%	20%
Finetune VGGNet-16 [50]	87.15±0.45	90.36±0.18
D-CNN with GoogleNet [51]	86.89±0.10	90.49±0.15
D-CNN with VGGNet 16 [51]	89.22±0.50	91.89±0.22
ResNet50 [52]	86.23±0.41	88.93±0.12
InceptionV3 [52]	85.46±0.33	87.73±0.43
LiG with sigmoid kernel [53]	90.19±0.11	93.21±0.12
LiG with RBF kernel [54]	90.23±0.13	93.25±0.12
Attention GAN [12]	88.06±0.19	90.58±0.23
ILOSR(ours)	**91.66±0.16**	**93.56±0.15**

NS and 1.33% higher than MES, which indicates that CHES is helpful to improve the accuracy of incremental learning. In addition, comparing the results of "CHES + our loss" with "CHES + CE loss" demonstrates the effectiveness of the proposed loss function.

Table III presents the effect of the number of selected exemplars in incremental learning. It can be observed that with the increase of the number of exemplars per class, the average accuracy over 45 classes and the training time per epoch are both rising. The accuracy tends to be relatively stable when the number of exemplars reaches 260. Therefore, we select 260 samples as replay exemplars for each class in the following experiments. Compared with the results of retraining all samples, the results of selecting 260 exemplars can reduce training time and storage consumption greatly without losing much accuracy, which presents the superiority of the proposed method in terms of efficiency and effectiveness.

C. Experiments for Offline RS Image Scene Classification

In this section, the performance of the proposed ILOSR method is evaluated in offline learning scenario on the NWPU-RESISC45 dataset. ILOSR is compared with eight state-of-the-art RS image scene classification methods, including Finetune VGGNet-16 [50], D-CNN with GoogleNet [51], D-CNN with VGGNet 16 [51], ResNet50 [52], InceptionV3 [52], LiG with sigmoid kernel [53], LiG with RBF kernel [54], and Attention GAN [12].

Table IV presents the comparison results of these methods in the case of 10% and 20% training ratios. The accuracy given in this table is averaged over 45 classes. It can be seen that our ILOSR method performs better than the other state-of-the-art methods. When the training ratio is 10%, the accuracy of our method reaches 91.66%, which is 1.43% higher than the best result provided by LiG with RBF kernel. When the training ratio is 20%, our method achieves 93.56% accuracy, which is 0.31% higher than the best result given by LiG with RBF kernel. Note that these compared methods are offline learning techniques, which suffers from catastrophic forgetting for incremental learning. Fig. 7 shows the t-distributed stochastic neighbor embedding (t-SNE) visualization result of our method in the feature space and some RS images are shown as examples. Each color represents one class. We can see that the intraclass features are compact and the interclass features are separated well, which should be attributed to the proposed new loss.

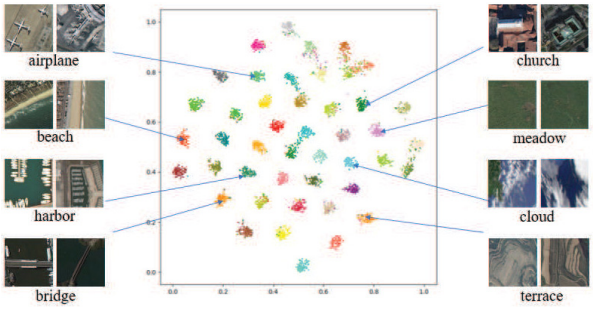


Fig. 7. Visualization of the feature space for 45 classes with different colors. Some RS images are shown as examples.

TABLE V
COMPARISON RESULTS OF DIFFERENT METHODS FOR INCREMENTAL LEARNING

Dataset	Number of classes	15	20	25	30	35	40	45
NWPU-RESISC45	Finetuning	0.9751	0.2407	0.1937	0.1601	0.1362	0.1255	0.1096
	LwF [36]	0.9767	0.5663	0.4382	0.3691	0.3400	0.3034	0.2761
	iCaRL [34]	0.9764	0.9507	0.9291	0.8999	0.8930	0.8847	0.8825
	BiC [55]	0.9790	0.9685	0.9514	0.9243	0.9100	0.9054	0.8945
	LUCIR [56]	0.9795	0.9692	0.9494	0.9404	0.9306	0.9264	0.9241
	ILOSR	**0.9809**	**0.9775**	**0.9680**	**0.9497**	**0.9442**	**0.9382**	**0.9347**
	Upperbound	**0.9819**	**0.9789**	**0.9694**	**0.9578**	**0.9477**	**0.9471**	**0.9466**
	Number of classes	6	7	8	9	10	11	12
SIRI-WHU	Finetuning	0.9908	0.1429	0.1250	0.1111	0.1000	0.0910	0.0833
	LwF [36]	0.9908	0.1429	0.1356	0.1105	0.0965	0.0914	0.0854
	iCaRL [34]	0.9902	0.9536	0.9644	0.9522	0.9555	0.9341	0.9242
	BiC [55]	0.9908	0.9645	0.9689	0.9544	0.9578	0.9431	0.9329
	LUCIR [56]	**0.9958**	0.9750	0.9718	0.9694	0.9575	0.9545	0.9521
	ILOSR	0.9946	**0.9914**	**0.9938**	**0.9872**	**0.9880**	**0.9768**	**0.9813**
	Upperbound	**0.9958**	**0.9964**	**0.9968**	**0.9972**	**0.9975**	**0.9841**	**0.9854**

D. Experiments for Incremental Learning

In this section, we evaluate the performance of ILOSR for incremental learning and compare ILOSR with several state-of-the-art incremental learning methods on the two datasets.

1) *NWPU-RESISC45:* The first 15 classes are set as the initial classes, and five classes are added each learning round according to the order in Fig. 6(a).
2) *SIRI-WHU*: The first six classes are set as the initial classes, and one class is added each time according to the order in Fig. 6(b).

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TABLE VI
EXPERIMENT SETTINGS FOR MULTICLASS OSR

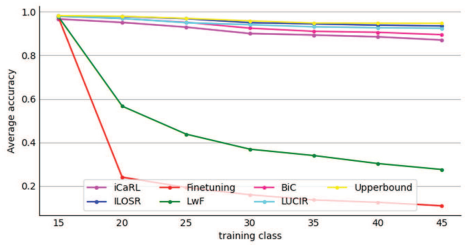
dataset		Known classes	Unknown classes	SUM
SIRI-WHU	Number of classes	6	6	12
	Number of train samples	7680	N/A	7680
	Number of test samples	240	240	480
	Label	0-5	-1	N/A
NWPU-RESISC45	Number of classes	15	30	45
	Number of train samples	8400	N/A	8400
	Number of test samples	2100	4200	6300
	Label	0-14	-1	N/A

The network model and related parameters are set the same as in Section IV-B. For comparing the performance of ILOSR, the following methods are considered.

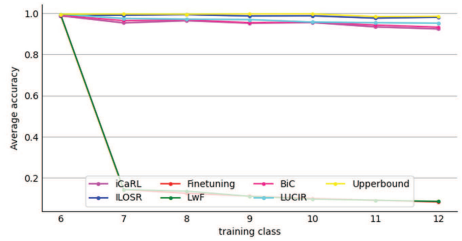
1) *Finetuning:* Train an initial network on the initial classes, which is the same as ILOSR, and then update the network for new incoming classes by finetuning the previous network. It does not take any measures to prevent catastrophic forgetting.
2) *LwF [36]:* The knowledge distillation loss is utilized to update the initial model and prevent catastrophic forgetting while learning new classes. It does not use the exemplar sets to replay data.
3) *iCaRL [34]:* Both the knowledge distillation loss and exemplar rehearsal are used to update the initial model and prevent forgetting.
4) *BiC [55]:* On the basis of iCaRL [34], a deviation correction layer by applying a linear model with a small validation set is added to address the imbalance issue. For the correction layer, there are 100 epochs, and the learning rate is set to 0.01 and reduces to 1/2 of the previous learning rate after 10, 30, 60, and 80 epochs.
5) *LUCIR [56]:* On the basis of iCaRL [34], three components, including cosine normalization, less-forget constraint, and interclass separation, are introduced to mitigate the adverse effects of the imbalance between old and new classes.
6) *Upperbound:* All the data of previously learned classes are stored and reused to train the current model. Thus, its performance can be regarded as the upper bound of incremental learning.

For iCaRL, BiC, and LUCIR, the number of exemplars for each class is also set as 260. For iCaRL, the nearest-mean-of-exemplars classification is used as the classifier. For the other methods, the fully connected layer of the network is used as a classifier. For all these methods, the learning rate and the batch size are set as 0.0001 and 32, respectively, and each training round runs 60 epochs.

Table V and Fig. 8 present the accuracy evolutions of different methods on the two datasets in the incremental learning process. It can be seen that all the methods get similar accuracies in the first learning round on 15 classes because their training sets are exactly the same initially. After the first round, Finetuning always gives the worst performance, which



(a)



(b)

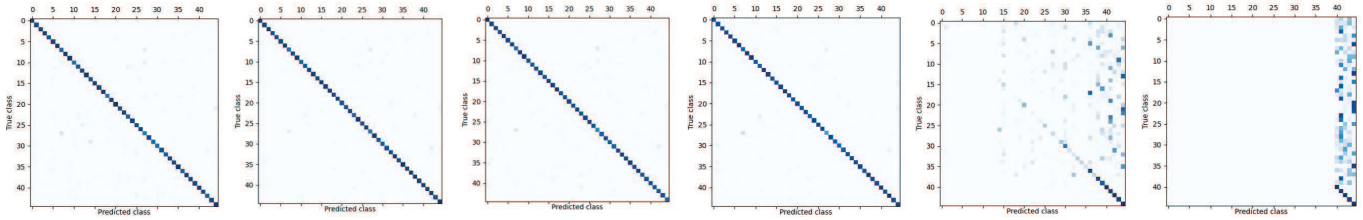
Fig. 8. Evolution of classification accuracy against the training classes in the incremental learning process. (a) NWPU-RESISC45 dataset. (b) SIRI-WHU dataset.

TABLE VII
COMPARISON RESULTS OF DIFFERENT METHODS FOR OSR

dataset	Methods	AUROC	Receiving Accuracy	Rejecting Accuracy	Average Accuracy
SIRI-WHU	SoftMax	0.8932	**0.8969**	0.7352	0.8161
	RPL	0.8591	0.8674	0.7925	0.8299
	ILOSR	**0.9077**	0.8590	**0.8314**	**0.8453**
NWPU-RESISC45	SoftMax	0.9173	**0.9358**	0.7283	0.8321
	RPL	0.9180	0.9258	0.7783	0.8521
	ILOSR	**0.9395**	0.9225	**0.8375**	**0.8855**

confirms that catastrophic forgetting is a critical problem for class-incremental learning. On NWPU-RESISC45, the accuracy of LwF increases 16% compared with that of Finetuning, but their accuracies are close on SIRI-WHU. This phenomenon shows that the distillation loss is useful to alleviate catastrophic forgetting, but its effect may be related to the number of new classes. The more the number of the new classes is, the more obvious its improvement may be. In addition, ILOSR, iCaRL, BiC, and LUCIR achieve significantly higher accuracy than Finetuning and LwF since they employ the exemplars replay. BiC gives better performance than iCaRL, which confirms the effect of deviation correction layer. LUCIR gets higher accuracy in most learning rounds due to the imbalance alleviation technique. In contrast, ILOSR outperforms all the above methods in each training round and its accuracy is very close to that of Upperbound. Compared with the best results of LUCIR, the accuracy of ILOSR increases by 1.06% on NWPU-RESISC45 and 2.92% on SIRI-WHU, which demonstrates the effectiveness of ILOSR for incremental learning.

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(a) (b) (c) (d) (e) (f)

Fig. 9. Confusion matrices of different methods on the NWPU-RESISC45 dataset. (a) ILOSR. (b) LUCIR. (c) BiC. (d) iCaRL. (e) LwF. (f) Finetuning.

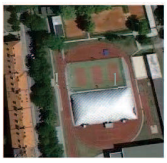
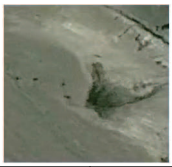
Images												
Methods	True label	Prediction	True label	Prediction	True label	Prediction	True label	Prediction	True label	Prediction	True label	Prediction
iCaRL	dense_ residential	commercial _area	Golf_ course	desert	ground_ trunk_field	baseball_ diamond	Inter- section	dense_ residential	Commer- cial_area	church	desert	beach
BiC	dense_ residential	Industrial _area	Golf_ course	desert	ground_ trunk_field	stadium	Inter- section	medium_ residential	Commer- cial_area	railway_ station	desert	mountain
LUCIR	dense_ residential	Mobile_ home_park	Golf_ course	mobile_ home_park	ground_ trunk_field	Tennis_ court	Inter- section	medium_ residential	Commer- cial_area	church	desert	mountain
ILOSR	dense_ residential	dense_ residential	Golf_ course	Golf _course	ground_ trunk_field	ground_ trunk_field	Inter- section	Inter- section	Commer- cial_area	church	desert	island

Fig. 10. Visualization classification results of six RS image examples from NWPU-RESISC45 dataset.

TABLE VIII
ACCURACY OF DIFFERENT METHODS FOR ILOSR ON THE NWPU-RESISC45 DATASET

Methods	Tasks	Set1	Set2	Set3	Set4	Set5	Set6	Set7	Avg.Acc
iCaRL+OSR (Baseline)	Task1(Set1)	**0.8902**	0.8083(U)	N/A	N/A	N/A	N/A	N/A	0.8492
	Task2(Task1+Set2)	0.5992	**0.9400**	**0.8291(U)**	N/A	N/A	N/A	N/A	0.7567
	Task3(Task2+Set3)	0.5761	**0.6997**	**0.9297**	**0.8277(U)**	N/A	N/A	N/A	0.7496
	Task4(Task3+Set4)	0.5000	0.6809	0.6614	**0.9029**	0.7371(U)	N/A	N/A	0.7026
	Task5(Task4+Set5)	0.5709	0.6791	0.6686	0.5563	**0.9089**	0.8951(U)	N/A	0.7708
	Task6(Task5+Set6)	0.5747	0.6486	0.6689	0.5634	**0.6291**	**0.8919**	0.8457(U)	0.7463
ILOSR	Task1(Set1)	0.8432	**0.8860(U)**	N/A	N/A	N/A	N/A	N/A	**0.8646**
	Task2(Task1+Set2)	**0.7770**	0.8109	0.7683(U)	N/A	N/A	N/A	N/A	**0.7769**
	Task3(Task2+Set3)	**0.7147**	0.6794	0.7660	0.8134(U)	N/A	N/A	N/A	**0.7667**
	Task4(Task3+Set4)	**0.7077**	**0.7003**	**0.7140**	0.7177	**0.9040(U)**	N/A	N/A	**0.8070**
	Task5(Task4+Set5)	**0.7025**	**0.7020**	**0.7111**	**0.5834**	0.7297	**0.9037(U)**	N/A	**0.7947**
	Task6(Task5+Set6)	**0.6802**	**0.6591**	**0.6891**	**0.5814**	0.5871	0.7866	**0.8794(U)**	**0.7717**

Fig. 9 shows the confusion matrices of different methods on the NWPU-RESISC45 dataset and gives further insight for the behaviors of these methods. It can be observed that the confusion matrices of ILOSR, LUCIR, BiC, and iCaRL are homogeneous over all classes, in terms of the diagonal entries (i.e., correct predictions) and off-diagonal entries (i.e., incorrect predictions). Thus, it illustrates that they do not have an intrinsic bias against classes that they encounter early or

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TABLE IX
ACCURACY OF DIFFERENT METHODS FOR ILOSR ON THE SIRI-WHU DATASET

Methods	Tasks	agricu-lture	comme-rcial	harbor	river	water	meadow	idle land	industrial	overpass	park	pond	residential	Avg. Acc
iCaRL+OSR (Baseline)	Task1(Initial 6 classes)	**1.0000**	**0.9300**	**0.9150**	**0.9700**	**0.9600**	**0.9300**	0.8700(U)	N/A	N/A	N/A	N/A	N/A	0.9104
	Task2(Task1+industrial)	0.6950	0.5900	0.5600	0.7050	0.8400	0.6900	**0.9900**	0.6000(U)	N/A	N/A	N/A	N/A	0.6621
	Task3(Task2+overpass)	0.7900	0.6050	0.4250	0.6250	0.7600	0.6500	0.6350	**0.9400**	**0.9300(U)**	N/A	N/A	N/A	0.8044
	Task4(Task3+park)	0.7050	0.5450	0.4650	0.4400	0.8150	0.5850	0.5350	0.6500	**0.9800**	**0.8800(U)**	N/A	N/A	0.6356
	Task5(Task4+pond)	0.7500	0.5850	**0.5800**	0.5800	0.8550	0.5750	**0.6800**	0.6950	**0.7500**	**0.9350**	0.8000(U)	N/A	0.7493
	Task6(Task5+residential)	0.7250	0.6200	0.5150	0.3950	0.7850	0.5900	0.6350	0.7200	0.6650	0.4950	**0.9500**	0.9050(U)	0.7750
ILOSR	Task1(Initial 6 classes)	0.9650	0.9100	0.8250	**0.9700**	0.9400	0.9200	**0.9800(U)**	N/A	N/A	N/A	N/A	N/A	**0.9508**
	Task2(Task1+industrial)	**0.9450**	**0.8250**	**0.5600**	**0.7850**	**0.9100**	**0.7950**	0.9300	**0.9800(U)**	N/A	N/A	N/A	N/A	**0.9007**
	Task3(Task2+overpass)	**0.9400**	**0.8100**	**0.6050**	**0.7100**	**0.8700**	**0.7900**	**0.7550**	0.8850	0.9250(U)	N/A	N/A	N/A	**0.8603**
	Task4(Task3+park)	**0.9150**	**0.7100**	**0.5500**	**0.7400**	**0.9200**	**0.7650**	**0.6850**	**0.7850**	0.8850	0.8250(U)	N/A	N/A	**0.7989**
	Task5(Task4+pond)	**0.8850**	**0.7100**	0.5250	**0.6350**	**0.9300**	**0.6600**	0.5800	**0.7250**	0.7350	0.7750	**0.8550(U)**	N/A	**0.7855**
	Task6(Task5+residential)	**0.8950**	**0.7300**	**0.6100**	**0.5650**	**0.9350**	**0.6500**	**0.6800**	**0.7400**	**0.8150**	**0.5850**	0.7950	**0.9700(U)**	**0.8486**

later. This means that they do not suffer from catastrophic forgetting due to the effect of the exemplars replay. In contrast, the confusion matrices of LwF and Finetuning look inhomogeneous. The two methods lead to intrinsic bias toward different classes and tend to predict samples as the latest learned classes. Compared with Finetuning, LwF performs better and alleviates this problem to a certain extent. Fig. 10 shows the visualization classification results of several RS scene images. For the first four images, the proposed ILOSR classifies them correctly, while iCaRL, BiC, and LUCIR predict them as other classes. For the last two images, all the methods misclassify them since they are very similar to other classes and easy to be confused.

E. Experiments for ILOSR

First, we evaluate the performance of ILOSR for multiclass OSR problems on the NWPU-RESISC45 and SIRI-WHU datasets. The experiments are set as follows.

1) *NWPUI-RESISC45:* The first 15 classes are set as the training classes, the rest 30 classes are set as the unknown classes, and the test set includes the whole 45 classes.
2) *SIRI-WHU:* The first six classes and the rest six classes are set as the training classes and unknown classes, respectively, and the test set includes the whole 12 classes.

Thus, the model should correctly classify the known classes and identify the unknown classes. The detailed experimental settings are shown in Table VI. The standard multiclass accuracy and the area under the receiver operating characteristic (AUROC) [44] are used as evaluation metrics to evaluate the performance of the closed-set classification and OSR. AUROC refers to the probability that a positive class sample is assigned a higher score than a negative class sample.

The network model and related parameters are set the same as the previous experiment. For ILOSR, we use one-class SVM with RBF kernel and parameters nu $= 0.005$ and $\gamma =$ "scale" for each class to establish a closed decision boundary in the feature space. For comparing the performance on multiclass OSR problems, we consider the following methods.

1) *SoftMax:* Train the 18-layer ResNet network based on the SoftMax function. The scores after softmax operation are used for open-set identification.
2) *RPL [45]:* Train the 18-layer ResNet network based on RPL. γ and λ are set as 0.5 and 0.1, respectively.

The number of reciprocal points is set as 1. Reciprocal points are initialized with ones and each margin is initialized with zeros. The probability values are used for open-set identification. The learning rate and the batch size of all the methods are set as 0.0001 and 32, respectively.

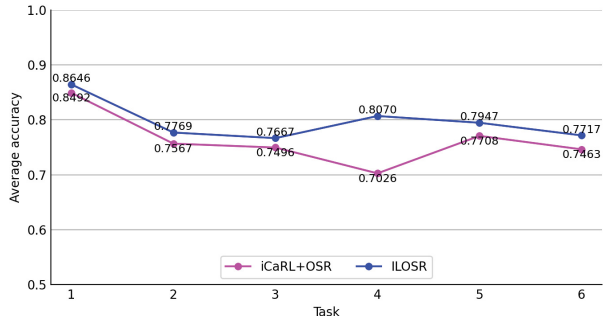
Table VII presents the comparison of experimental results on the NWPU-RESISC45 and SIRI-WHU datasets for OSR. It can be seen that ILOSR outperforms the other methods in terms of AUROC and rejecting accuracy, which demonstrates that ILOSR learns better scores for identifying the unknown classes. Since one-class SVM establishes the closed decision boundaries in the feature space, ILOSR can better manage the known sample space and the open space than the other methods. Besides, the features learned by ILOSR are more discriminative than the other methods, which is helpful to identify the unknown classes.

In Section IV-E1, we will evaluate the performance of ILOSR in the scenario of ILOSR on NWPU-RESISC45 and SIRI-WHU. The experiment settings of the two datasets are the same as those in Section IV-D. The network model and parameters are set the same as the previous experiments.

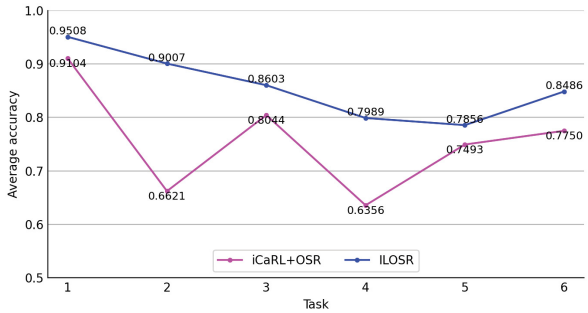
1) Baseline: Note that there is no agreed upon baseline for ILOSR so far. Therefore, we set a competitive baseline, named iCaRL+OSR, based on the well-trained iCaRL model and one-class SVM. The SVM is used to establish the closed boundaries and identify the unknown classes in feature space. The network model and the parameters are the same as those of ILOSR.

Tables VIII and IX show the results of different methods on the SIRI-WHU and NWPU-RESISC45 datasets, respectively.

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(a)



(b)

Fig. 11. Evolution of average accuracy against the encountered tasks in the learning processing. (a) NWPU-RESISC45 dataset. (b) SIRI-WHU dataset.

Note that "U" denotes the unknown classes. For instance, in Table IX, the row of "Task1" presents the classification accuracy of each known class and the unknown class (idle land). The last column of "Avg. Acc" is the average value of the accuracy on the known and unknown classes. After identifying the unknown classes and labeling them as the new known classes, the current model is updated to incrementally learn these new classes by using exemplars of the old classes and the new classes. The new model is utilized to test the data of Task 2 and the separated average accuracies of the known classes and the unknown class (industrial) are shown in the row of "Task2." The experiments of the rest rows in Table IX are carried out in the same manner. In Table VIII, due to the large number of classes, the accuracy of each class cannot be displayed. Thus, we divide the classes into six sets according to the training round with "Set1," including 15 classes, while the rest of each set includes five classes. From Tables VIII and IX, it can be seen that ILOSR gives higher accuracy than the baseline for ILOSR in most cases. It demonstrates that ILOSR can effectively detect the unknown classes and continually learn the new classes while maintaining reasonable accuracies for the known classes without catastrophic forgetting in the learning process. Notice that the accuracies of recognizing the unknown classes and classifying the known classes in different tasks are not monotonically decreasing. This is because the similarity between the known classes and different unknown classes is variable and the similar classes may partially overlap in the feature space. Fig. 11 presents the average accuracies of the known and unknown classes on the two datasets. We can observe that ILOSR outperforms iCaRL+OSR and the average accuracy of the baseline fluctuates obviously, while that of ILOSR is relatively stable, which should ascribe to the more representative exemplars and separable class feature given by ILOSR.

V. Conclusion

In this article, an ILOSR framework has been proposed for RS image scene classification in the open and dynamic environment. First, a new loss function based on prototype learning and uncertainty measurement is proposed for the OSR problem. This technique improves the interclass discrimination and intraclass compactness of deep features, which makes the learned feature space better separate the known and unknown classes. In addition, a controllable CHES method is introduced to alleviate catastrophic forgetting in incremental learning, which can reduce the training time and memory footprint. The proposed ILOSR can identify the unknown classes accurately and learn the new classes incrementally. Experiments on real RS datasets demonstrate that ILOSR can achieve the state-of-the-art performance in the separate and integral experiments of incremental learning and OSR problems. ILOSR is a foundation framework to continually recognize unknown classes and learn new data or classes from large-scale and dynamic RS data streams, whose efficiency and effectiveness have been verified by the experimental results.

The proposed ILOSR method still relies on learning new classes from a large amount of data and requires to save and replay some samples of previously learned classes to overcome catastrophic forgetting. In some applications, usually, only a few number of examples belonging to unknown classes can be acquired, and the samples of previous classes may be inaccessible due to memory constraints or data legality issues. In our future work, we will extend the proposed framework to few-shot and data-free incremental learning problems to learn effectively from a small number of samples without replaying previously seen data.

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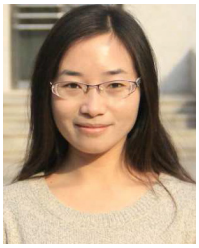
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